

Hematopoietic stem cell transplantation for HIV cure

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The apparent cure of an HIV-infected person following hematopoietic stem cell transplantation (HSCT) from an allogeneic donor homozygous for the *CCR5*Δ32 mutation has stimulated the search for strategies to eradicate HIV or to induce long-term remission without requiring ongoing antiretroviral therapy. A variety of approaches, including allogeneic HSCT from *CCR5*-deficient donors and autologous transplantation of genetically modified hematopoietic stem cells, are currently under investigation. This Review covers the experience with HSCT in HIV infection to date and provides a survey of ongoing work in the field. The challenges of developing HSCT for HIV cure in the context of safe, effective, and convenient once-daily antiretroviral therapy are also discussed.

Introduction

The advent of potent combination antiretroviral therapy (ART) has led to a dramatic decrease in the incidence of AIDS and AIDS-related mortality worldwide. For most patients, full suppression of HIV-1 replication can be achieved by once-daily administration of an ART regimen available as a fixed-dose combination that is safe, convenient, and well tolerated. Five single-tablet regimens are currently approved by the US FDA for the initial treatment of HIV-1 infection, and several more are in phase III clinical trials (e.g., NCT02269917; NCT01797445; NCT02345226). Some studies suggest that, with early initiation of ART, patients with HIV-1 infection may live essentially normal life spans (1), converting what was once a uniformly fatal viral infection into a chronic disease that is manageable with appropriate medical treatment.

Although effective at restoring immune function and prolonging life, ART does not eliminate HIV-1, which persists as a latent infection in resting memory CD4⁺ T cells and possibly in cells of monocyte origin; treatment, therefore, must be administered throughout a patient's life. Despite the advances in HIV therapeutics of the last 30 years, concerns remain regarding the long-term safety of decades of ART, the burden of daily adherence, and the costs of providing lifelong ART on a global scale. In addition, higher-than-normal levels of immune activation persist in patients with full suppression of viral replication, and this activation is associated with an increased risk of end-organ disease, including myocardial infarction and stroke (2–4). A treatment that led to durable drug-free remission or eradication (cure) of HIV-1 could reduce the burden, cost, toxicities, and stigma associated with long-term ART and might lower immune activation and the associated risk of non-AIDS clinical events. The search for a cure therefore remains a high priority for clinicians, investigators, and patients.

As reviewed elsewhere in this issue (5), little or no transcription of proviral DNA occurs in resting CD4⁺ T cells that are latently infected with HIV-1. In the absence of any expression of viral pro-

teins, these cells evade detection and destruction by the innate and adaptive immune systems. If latently infected cells could be eliminated (e.g., by ablative chemotherapy) and replaced through hematopoietic stem cell transplantation (HSCT) by uninfected cells or, ideally, cells intrinsically resistant to HIV-1 infection, cure might be achieved. This approach has attracted the interest of investigators for many years but, to date, has resulted in only a single successful outcome (6).

Early experience with HSCT in HIV-infected patients

The history of HSCT in HIV-infected patients has been reviewed by Hütter and Zaia (7). Early attempts to apply HSCT as an approach to immune reconstitution in patients with AIDS or as treatment for hematologic malignancies met with little success (8–10). Unsurprisingly, in the absence of effective ART, HSCT had little impact on the course of HIV disease and most patients died of progressive immunodeficiency or recurrent leukemia or lymphoma. In one case report, a patient with refractory lymphoma received an allogeneic HSCT from a matched, unrelated donor following conditioning with cyclophosphamide and total-body irradiation (TBI), along with zidovudine, which had recently become available (11). Engraftment occurred on day 17, and the patient subsequently demonstrated complete chimerism. Virus was undetectable by culture or PCR of peripheral blood mononuclear cells (PBMC) beginning at day 32 after transplant. Unfortunately, the patient died of recurrent lymphoma at day 47, but HIV-1 was undetectable by culture or PCR of a variety of tissues obtained at autopsy. A similar outcome was reported in another patient who received an allogeneic HSCT and zidovudine; HIV-1 became undetectable by PCR in blood following engraftment, but the patient succumbed to graft-versus-host diseases (GVHD) (R. Saral and H.K. Holland, cited in ref. 7). In another case, a 25-year old woman with AIDS who received an allogeneic HSCT from a matched, unrelated donor after conditioning with busulfan and cyclophosphamide in the setting of ART with zidovudine and IFN-α2 survived for 10 months before succumbing to adult respiratory distress syndrome (12); tissues obtained at autopsy were negative for HIV-1 by PCR.

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